

Solar Neutrino Mini-Project

Team name

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1 Physics question

State the concrete question your team answered.

Use natural units, $\hbar = c = 1$, for analytic formulas. State practical table units explicitly when quoting numerical inputs and plot axes.

2 Model

Write the minimum equations used in the project. Explain which parts are physical inputs and which parts are toy approximations.

3 Input data

List the data files and reference values used:

- solar flux table;
- ^8B -spectrum;
- electron density profile;
- detector response parameters.

4 Code and numerical method

Describe the notebook and the key functions from `src/solar_neutrino`.

Include one short code fragment only if it clarifies the method.

5 Results

Insert the main figure and the numerical result:

```
best-fit normalization =  
best-fit effective  $P_{ee}$  =  
confidence interval =
```

6 Discussion of approximations

State explicitly:

- fluxes are inputs, not a solved solar model;
- ^8B -spectrum may be a toy beta-like shape;
- detector response is schematic;
- backgrounds and systematics are simplified or absent.

7 Conclusion

Write a short scientific conclusion that follows from the plotted result.

8 References